

Exam 1: NE533: Nuclear Fuel Performance

Show all work. Label question number in your response. Pay attention to units. Point values correspond to expected depth of response.

1. (12 pts) UN is a fuel being considered for use in light water reactors. Consider UN to have an enrichment of 19% and a density of 14.3 g/cm^3 . Assume the fission cross section is 587 barns. Nitrogen mass number=14.

What is the heat generation rate, given a neutron flux of $4 \times 10^{12} \text{ n/cm}^2\text{-s}$? What enrichment of UO_2 would be required to obtain the same heat generation rate? UO_2 density is 10.97 g/cc .

2. (18 pts) Calculate the temperature at the radial midpoint of the cladding and at $r=0.1 \text{ cm}$. Assume gap is all He.

Cladding k : 0.18 W/cm-K ; Fuel k : 0.04 W/cm-K ; Coolant $h = 1.5 \text{ W/cm}^2\text{-K}$; $Q = 350 \text{ W/cm}^3$; $R_F=0.45 \text{ cm}$; $t_g=80 \text{ }\mu\text{m}$; $t_{\text{clad}}=0.06 \text{ cm}$; $T_{\text{cool}} = 540 \text{ K}$

3. (8 pts) Given a rod of 3.6 m in length, $\text{LHR}^0 = 400 \text{ W/cm}$, and $\gamma=1.25$:

What is the LHR at $z=2.1 \text{ m}$? What is the coolant outlet temperature? Assume water: $C_p = 4200 \text{ J/kg-K}$, $\dot{m} = 0.1 \text{ kg/s-rod}$, inlet temperature = 580 K .

4. (6 pts) By how much does the thermal conductivity of UO_2 degrade at a burnup of 5% FIMA? Assume an average fuel temperature of 1200 K .
5. (6 pts) What assumptions to the heat conduction equation were required to derive the analytical form of the temperature profile?
6. (5 pts) What compound is utilized in the enrichment process? Describe the centrifuge-based enrichment of U, including why it works.
7. (8 pts) Discuss the departure from nucleate boiling ratio and the critical heat flux.
8. (4 pts) What is smear density? Why is a reduced smear density necessary?
9. (4 pts) What are two sources of thermal conductivity degradation in UO_2 ?
10. (6 pts) What is the role of cladding? Why is Zr suitable as a cladding material?
11. (4 pts) What does the "fuel system" consist of?
12. (4 pts) What are the three aspects that I define as constituting fuel performance?
13. (6 pts) What are the three ways that space is discretized for numerical solutions? Name a strength or weakness of one of these types.
14. (5 pts) Name two positive and two negative aspects of UO_2 fuel.
15. (4 pts) What is the conceptual difference between implicit and explicit time integration?

Equations for NE533

$$\begin{aligned}
 Q &= E_f \times N_f \times \sigma_f \times \phi \\
 \rho c_p \frac{\partial T}{\partial t} &= \nabla \cdot (k \nabla T) + Q \\
 LHR &= \pi R^2 Q_{av} \\
 T(r) &= \frac{Q(R^2 - r^2)}{4k} + T_s \\
 T_0 - T_s &= \frac{Q}{4k} R_f^2 = \frac{LHR}{4\pi k} \\
 T_s - T_{IC} &= \frac{Q}{2h_{gap}} R_f = \frac{LHR}{2\pi R_f h_{gap}} \\
 T_{IC} - T_{OC} &= \frac{Q}{2h_{clad}} R_f = \frac{LHR}{2\pi R_f h_{clad}} \\
 T_{OC} - T_{cool} &= \frac{Q}{2h_{cool}} R_f = \frac{LHR}{2\pi R_f h_{cool}} \\
 h_{gap/clad} &= \frac{k_{gap/clad}}{t_{gap/clad}} \\
 k_{gap}(He) &= 16 \times 10^{-6} \times T^{0.79} \\
 k_{gap}(Xe) &= 0.7 \times 10^{-6} \times T^{0.79} \\
 k_{gap}(y) &= (1 - y) \times k_{He} + y \times k_{Xe} \\
 LHR \left(\frac{z}{Z_0} \right) &= LHR^0 \cos \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \\
 T_{cool}(z) - T_{cool}^{in} &= \frac{2\gamma}{\pi} \frac{Z_0 \times LHR^0}{\dot{m} C_p} \left\{ \sin \frac{\pi}{2\gamma} + \sin \left[\frac{\pi}{2\gamma} \left(\frac{z}{Z_0} - 1 \right) \right] \right\} \\
 k_{ox} &= \frac{1}{A + BT} \text{ (W/cmK)} \\
 \frac{1}{B} \ln \left(\frac{A + BT_0}{A + BT_s} \right) &= \frac{LHR}{4\pi} \\
 A &= 3.8 + 200 \times FIMA \text{ (cmK/W)} \\
 B &= 0.0217 \text{ (cm/W)}
 \end{aligned}$$